

AEP and Source Coding

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EE 708: Information Theory and Coding

Concentration inequalities - part 1

Markov inequality

For a non-negative r.v., and $a > 0$,

$$\Pr(X \geq a) \leq \frac{E(X)}{a}.$$

Proof:

$$\begin{aligned} E(X) &= \sum_{x \in \mathcal{X}} x \cdot \Pr(X = x) \\ &\geq \sum_{x \geq a} a \cdot \Pr(X = x) \\ &= a \cdot \sum_{x \geq a} \Pr(X = x) \\ &= a \cdot \Pr(X \geq a) \end{aligned}$$

Chernoff bound

Chernoff bound

For a r.v. X and any $t > 0$,

$$\Pr(X \geq a) = \Pr(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}.$$

Proof: Follows by using Markov inequality on e^{tX} .

Chernoff bound: sum of i.i.d. random variables

Chernoff bound

For i.i.d. zero-mean random variables $X_1, X_2, \dots, X_n$, define $Y = \frac{1}{n} \sum_i X_i$.

$$\Pr(Y \geq a) = e^{-n \cdot \sup_{t>0} (ta - \ln E[e^{tX_1}])}$$

Proof:

$$\begin{aligned} \Pr(Y \geq a) &= \Pr(\sum X_i \geq na) && \leq \inf_{t>0} \frac{E[e^{t \sum X_i}]}{e^{tna}} \\ &= \inf_{t>0} e^{-tna} \prod_i E[e^{tX_i}] \\ &= \inf_{t>0} e^{-tna} (E[e^{tX_1}])^n \\ &= e^{-n \cdot \sup_{t>0} (ta - \ln E[e^{tX_1}])} \end{aligned}$$

Hoeffding bound

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables, taking values in $\{0, 1\}$. Denote $Y = (1/n) \sum_{i=1}^n X_i$. Let $p = \Pr(X = 1)$, and $\epsilon > 0$. Then

$$\Pr(Y \geq p + \epsilon) \leq 2^{-nD(p+\epsilon||p)}.$$

$$\Pr(Y \leq p - \epsilon) \leq 2^{-nD(p-\epsilon||p)}.$$

Note: $D(q||p)$ denotes $D(\text{Bern}(q)||\text{Bern}(p))$.

For $\lambda \in (0, 1)$, and large n ,

$$\binom{n}{n\lambda} \approx 2^{nH(\lambda)}$$

Proof: Sterling's approximation:

$$\begin{aligned} \log_2 n! &= n \log_2 n - n - (1/2) \log_2 n + (1/2) \log_2(2\pi) + o(1) \\ &= n \log_2 n - n + o(n) \end{aligned}$$

$$\begin{aligned} \binom{n}{n\lambda} &= \frac{n!}{n\lambda!(n - n\lambda)!} \\ &= \frac{2^{n \log_2 n - n}}{2^{n\lambda \log_2(n\lambda) - n\lambda} 2^{n(1-\lambda) \log_2 n(1-\lambda) - n(1-\lambda)}} \cdot 2^{o(n)} \\ &= 2^{n[-\lambda \log_2 \lambda - (1-\lambda) \log_2(1-\lambda) + o(1)]} \\ &= 2^{n(H(\lambda) + o(1))} \end{aligned}$$

Volume of a Ball

Hamming distance between $\mathbf{x}, \mathbf{y} \in \{0, 1\}^n$:

$$d_H(\mathbf{x}, \mathbf{y}) := |\{i | x_i \neq y_i\}|,$$

the number of components of $\mathbf{x}, \mathbf{y}$ that are different.

Ball of radius $r \leq n$ around $\mathbf{x}$:

$$B_r(\mathbf{x}) := \{\mathbf{y} | d_H(\mathbf{x}, \mathbf{y}) \leq r\}$$

Volume of a ball (for $r < n/2$)

$$\begin{aligned} |B_r(\mathbf{x})| &= \sum_{i=0}^r \binom{n}{i} \\ &\approx \sum_{i=0}^r 2^{n(H(i/n))} \\ &\approx 2^{n(H(r/n))} \end{aligned}$$

Hoeffding bound: Proof

$$Pr(Y \geq p + \epsilon) \leq 2^{-nD(p+\epsilon||p)}.$$

Note that, for $i = n\lambda$ (when i is an integer), the probability that the vector $(X_1, \dots, X_n)$ has i 'one's' is given by

$$\begin{aligned} \binom{n}{n\lambda} p^{n\lambda} (1-p)^{n-n\lambda} &\approx 2^{nH(\lambda)} \cdot 2^{n\lambda \log_2 p} \cdot 2^{n(1-\lambda) \log_2 (1-p)} \\ &= 2^{-n \left[\lambda \log_2 \frac{\lambda}{p} + (1-\lambda) \log_2 \frac{(1-\lambda)}{(1-p)} \right]} \\ &= 2^{-nD(\lambda||p)} \end{aligned}$$

This is the probability that $\mathbf{X}$ has an empirical distribution $Bern(\lambda = \frac{i}{n})$.

Hoeffding bound: Proof (contd.)

$D(\lambda||p)$ is monotonically increasing for $\lambda > p$. So

$$\begin{aligned} Pr(Y \geq p + \epsilon) &= \sum_{i=\lceil n(p+\epsilon) \rceil}^n \binom{n}{i} p^i (1-p)^{n-i} \\ &= \sum_{i=\lceil n(p+\epsilon) \rceil}^n 2^{nD(\frac{i}{n}||p)} \\ &\leq n \cdot 2^{-nD(p+\epsilon||p)} \approx 2^{-nD(p+\epsilon||p)}. \end{aligned}$$

$$Pr(Y \geq p + \epsilon) \leq 2^{-n\epsilon^2 \log_2 e/2}$$

Pinsker's inequality:

$$\begin{aligned} D(p||q) &\geq 2d_{TV}(p, q)^2 \log_2 e \\ \Rightarrow D(p||q) &\geq \frac{\log_2 e}{2} \|p - q\|_1^2 \end{aligned}$$

Sanov's theorem

Types

Type of a vector

For any $a \in \mathcal{X}$, let $N_{\mathbf{x}}(a)$ denote the number of a in the vector $\mathbf{x} \in \mathcal{X}^n$, and $T_{\mathbf{x}}(a) = N_{\mathbf{x}}(a)/n$.

The vector $(T_{\mathbf{x}}(a))_{a \in \mathcal{X}}$ is called the **type** (or “empirical distribution” or “composition”) of $\mathbf{x}$.

For binary vectors, $\sum x_i$ determines the type.

Note:

- It is a distribution.
- It has components of the form k/n .

Question: How many possible types exist for a given length n ?

$$\text{Number of types} \leq (n+1)^{|\mathcal{X}|}$$

Generalization

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with pmf $\sim p$ over the alphabet $\mathcal{X}$. Let q be a valid type for n -length vectors in $\mathcal{X}$. Then

$$Pr(T_{\mathbf{X}} = q) \approx 2^{-nD(q||p)}.$$

Proof is left as an exercise.

Step 1: First prove that the number of vectors of type q

$$= 2^{n(H(q)+o(1))}$$

Step 2: Number of vectors of type $q = ?$

Sanov's theorem

Sanov's theorem

Let $X_1, X_2, \dots, X_n$ be i.i.d. $\sim Q(x)$. Let E be a subset of all probability distributions no $\mathcal{X}$. Then

$$Pr(T_{\mathbf{X}} \in E) \leq (n+1)^{|\mathcal{X}|} 2^{-nD(P^*||Q)}$$

where $P^* = \arg \min_{P \in E} D(P||Q)$

Types and strongly typical sequences

Strongly typical set

Definition: Type of a vector

For any $a \in \mathcal{X}$, let $N_{\mathbf{x}}(a)$ denote the number of a in the vector $\mathbf{x} \in \mathcal{X}^n$, and $f_{\mathbf{x}}(a) = N_{\mathbf{x}}(a)/n$. The vector $(f_{\mathbf{x}}(a))_{a \in \mathcal{X}}$ is called the **type** of $\mathbf{x}$.

Note:

- Type is a distribution.
- It has components of the form k/n .

Question: How many possible types exist for a given length n ?

$$\text{Number of types} \leq (n+1)^{|\mathcal{X}|}$$

Strongly typical sequence

Definition: Strongly typical vector

The vector $\mathbf{x}$ is called strongly ϵ -typical if $|p(a) - f_{\mathbf{x}}(a)| < \epsilon p(a)$.

For a strongly typical sequence,

$$\begin{aligned} p(\mathbf{x}) &= \prod_{a \in \mathcal{X}} p(a)^{nf_{\mathbf{x}}(a)} &&= 2^{n \cdot \sum_{a \in \mathcal{X}} f_{\mathbf{x}}(a) \log_2 p(a)} \\ &\in 2^{n \cdot \sum_{a \in \mathcal{X}} (1 \pm \epsilon) p(a) \log_2 p(a)} \\ &= 2^{-nH(X)(1 \pm \epsilon)} \\ &\subseteq 2^{-n(H(X) \pm \epsilon')} \end{aligned}$$

Typical set: $T_{\epsilon}^{(n)} :=$ set of all ϵ -typical vectors of length n

Typical average lemma

Suppose $\mathbf{x}$ is a typical sequence. Then for any nonnegative function $g(x)$ on $\mathcal{X}$,

$$(1 - \epsilon)E(g(X)) \leq \frac{1}{n} \sum_{i=1}^n g(x_i) \leq (1 + \epsilon)E(g(X))$$

Proof:

$$\begin{aligned} \left| \frac{1}{n} \sum_{i=1}^n g(x_i) - E(g(X)) \right| &= \left| T_{\mathbf{x}}(x)g(x) - \sum_x p(x)g(x) \right| \\ &\leq \sum_x \epsilon p(x)g(x) \\ &= \epsilon \cdot E(g(X)) \end{aligned}$$

Question: What does this lemma imply if we take $g(\cdot) := -\log p(\cdot)$?

Probability of the typical set

Assume $\min_{a \in \mathcal{X}} p(a) = p_{\min} > 0$ and $0 < \epsilon < p_{\min}$.

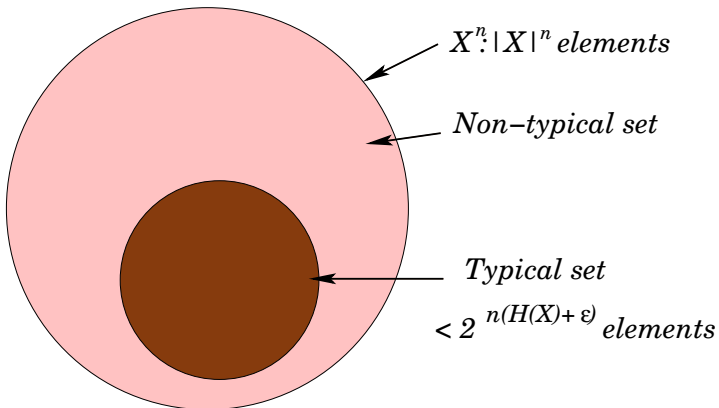
Probability of the typical set

Prove using Hoeffding bound that

$$\Pr(A_{\epsilon}^{(n)}) \geq 1 - 2|\mathcal{X}|2^{-n\epsilon'^2 \log e/2}$$

where $\epsilon' = \epsilon/H(X)$.

The big picture



“The smallest high-probability set”

Caution

- Typical sequences are not necessarily high-probability sequences.
- E.g., take $X_1, X_2, \dots, X_n \sim \text{Ber}(3/4)$. The highest probability sequence is *all-ones*, which is not typical.

Weakly typical sequences

WLLN: Convergence in Probability

Definition: convergence in probability

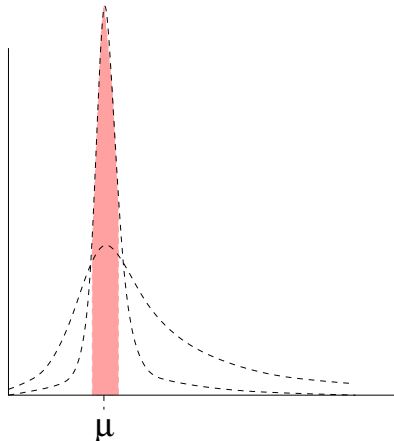
For every $\epsilon > 0$, $Pr\{|X_n - \mu| > \epsilon\} \rightarrow 0$

Weak law of large numbers (WLLN)

$X_1, X_2, X_3, \dots$, i.i.d. with mean μ .

$Y_n = \frac{1}{n} \sum_{i=1}^n X_i$ - partial average

$Y_n \rightarrow \mu$ in probability



Asymptotic equipartition property

Theorem (AEP)

If $X_1, X_2, \dots$ are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n} \log p(X_1, X_2, \dots, X_n) \rightarrow H(X) \text{ in probability}$$

Proof:

If X_i are i.i.d., then so are $\log p(X_i)$.

$$\begin{aligned} -\frac{1}{n} \log p(X_1, X_2, \dots, X_n) &= -\frac{1}{n} \sum_i \log p(X_i) \\ &\rightarrow -E \log p(X) \text{ in probability} \\ &= H(X) \end{aligned}$$



Asymptotic equipartition property

AEP implies:

$$\Pr \left\{ \left| -\frac{1}{n} \log p(X_1, X_2, \dots, X_n) - H(X) \right| > \epsilon \right\} \rightarrow 0$$

So the complement of the above event has probability:

$$\begin{aligned} & \Pr \{ | -\log p(X_1, X_2, \dots, X_n) - nH(X) | \leq n\epsilon \} \\ = & \Pr \{ n(H(X) - \epsilon) \leq -\log p(X_1, X_2, \dots, X_n) \leq n(H(X) + \epsilon) \} \\ = & \Pr \{ -n(H(X) - \epsilon) \geq \log p(X_1, X_2, \dots, X_n) \geq -n(H(X) + \epsilon) \} \\ = & \Pr \left\{ 2^{-n(H(X) - \epsilon)} \geq p(X_1, X_2, \dots, X_n) \geq 2^{-n(H(X) + \epsilon)} \right\} \\ \rightarrow & 1 \end{aligned}$$

(Weakly) Typical set

Definition

The typical set $A_\epsilon^{(n)}$ with respect to a distribution $p(x)$ is the set of all sequences $(x_1, x_2, \dots, x_n) \in \mathcal{X}^n$ with the property

$$2^{-n(H(X)+\epsilon)} \geq p(X_1, X_2, \dots, X_n) \geq 2^{-n(H(X)-\epsilon)}$$

Relation: Strongly vs. Weakly typical sequences

Recall that for a strongly typical sequence,

$$\begin{aligned}
 p(\mathbf{x}) &= \prod_{a \in \mathcal{X}} p(a)^{nf_{\mathbf{x}}(a)} &= 2^{n \cdot \sum_{a \in \mathcal{X}} f_{\mathbf{x}}(a) \log_2 p(a)} \\
 &\in 2^{n \cdot \sum_{a \in \mathcal{X}} (1 \pm \epsilon) p(a) \log_2 p(a)} \\
 &= 2^{-nH(X)(1 \pm \epsilon)} \\
 &\subseteq 2^{-n(H(X) \pm \epsilon')}
 \end{aligned}$$

A strongly ϵ -typical sequence is also weakly $g(\epsilon)$ -typical sequence (for some $g(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$).

Exercise: What is the function $g(\epsilon)$ above.

- *A weakly typical sequence may not be strongly typical.*
Example: $\mathcal{X} = \{0, 1\}$, prob. distribution = $\{0.5, 0.5\}$. Any sequence is weakly typical but not necessarily strongly typical.
- Weakly typical sequences may not 'look' typical. Take $Ber(1/2)$ - all sequences are weakly typical.

Properties of the typical set

Properties of the typical set

Properties

1. If $(x_1, x_2, \dots, x_n) \in A_\epsilon^{(n)}$, then

$$H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1, x_2, \dots, x_n) \leq H(X) + \epsilon.$$
2. For any $\delta > 0$, $\Pr \left\{ A_\epsilon^{(n)} \right\} > 1 - \delta$ for sufficiently large n .

In fact, we know that $\Pr(A_\epsilon^{(n)}) \geq 1 - 2|\mathcal{X}|2^{-n\epsilon^2 \log e/2}$.

3. $\left| A_\epsilon^{(n)} \right| \leq 2^{n(H(X)+\epsilon)}$
4. For any $\delta > 0$, $\left| A_\epsilon^{(n)} \right| \geq (1 - \delta)2^{n(H(X)-\epsilon)}$ for sufficiently large n .

Proof of the Properties

1. Follows from the definition.
2. Follows from AEP.
- 3.

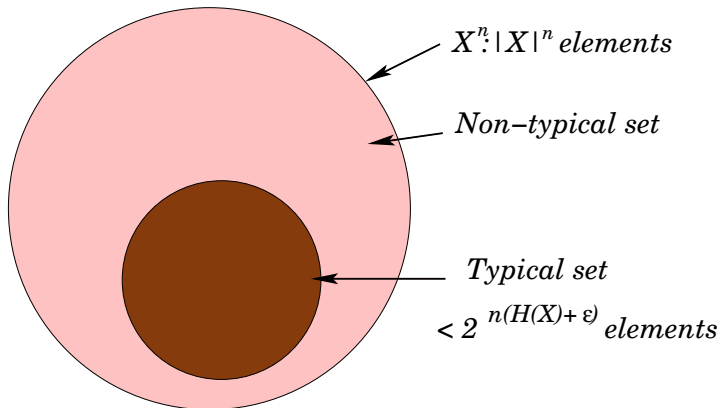
$$\begin{aligned}
 1 &= \sum_{\mathbf{x} \in \mathcal{X}^n} p(\mathbf{x}) \\
 &\geq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} p(\mathbf{x}) \\
 &\geq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} 2^{-n(H(X)+\epsilon)} \\
 &= 2^{-n(H(X)+\epsilon)} |A_\epsilon^{(n)}| \\
 \Leftrightarrow |A_\epsilon^{(n)}| &\leq 2^{n(H(X)+\epsilon)}
 \end{aligned}$$

Proof of the Properties

4. For sufficiently large n , $Pr\{A_\epsilon^{(n)}\} > 1 - \delta$. So

$$\begin{aligned}
 1 - \delta &< Pr\{A_\epsilon^{(n)}\} \\
 &\leq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} 2^{-n(H(X) - \epsilon)} \\
 &= 2^{-n(H(X) - \epsilon)} |A_\epsilon^{(n)}| \\
 \Leftrightarrow |A_\epsilon^{(n)}| &\geq (1 - \delta) 2^{n(H(X) - \epsilon)}
 \end{aligned}$$

The big picture



“The smallest high-probability set”

Recall

- Typical sequences are not necessarily high-probability sequences.
- E.g., take $X_1, X_2, \dots, X_n \sim \text{Ber}(3/4)$. The highest probability sequence is *all-ones*, which is not typical.
- Weakly typical sequences may not 'look' typical. Take $\text{Ber}(1/2)$ - all sequences are weakly typical.

AEP and Source Coding

Source coding

Zero Error, Variable length

Construct a prefix code

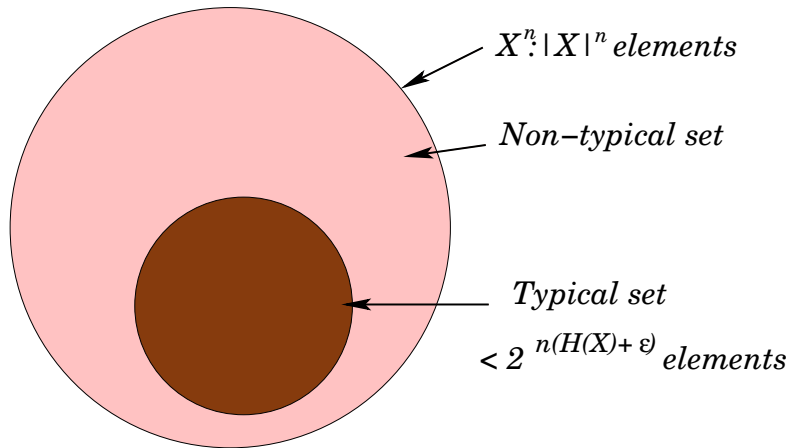
- Deterministic construction; achieves $H(X)$ - short codewords for high probability sequences
- Deterministic construction - short codewords for typical sequences

ϵ -Error, fixed length

Ensure that the typical sequences are assigned distinct codewords

- Deterministic construction
- Random coding

The big picture



Source coding: Zero Error, Variable length

Codes for the typical set

0 00 00
0 00 01
:
0 11 11

$$\begin{aligned}\text{Codelength} &\leq 1 + \log(2^{n(H(X)+\epsilon)}) \\ &\leq 2 + n(H(X) + \epsilon)\end{aligned}$$

Codes for the non-typical set

1 00 00
1 00 01
:
1 11 11

$$\begin{aligned}\text{Codelength} \\ &\leq 2 + \log(|\mathcal{X}|^n) \leq n \log |\mathcal{X}| + 2\end{aligned}$$

Source coding: Zero Error, Variable length

Average codelength

$$\begin{aligned}
 &= \sum_{\mathbf{x} \in \mathcal{X}^n} p(\mathbf{x}) l(\mathbf{x}) \\
 &= \sum_{\mathbf{x} \in A_\epsilon^{(n)}} p(\mathbf{x}) l(\mathbf{x}) + \sum_{\mathbf{x} \in A_\epsilon^{(n)c}} p(\mathbf{x}) l(\mathbf{x}) \\
 &\leq \sum_{\mathbf{x} \in A_\epsilon^{(n)}} p(\mathbf{x}) (n(H + \epsilon) + 2) + \sum_{\mathbf{x} \in A_\epsilon^{(n)c}} p(\mathbf{x}) (n \log |\mathcal{X}| + 2) \\
 &= Pr\{A_\epsilon^{(n)}\} (n(H + \epsilon) + 2) + Pr\{A_\epsilon^{(n)c}\} (n \log |\mathcal{X}| + 2) \\
 &\leq n(H + \epsilon) + \epsilon n(\log |\mathcal{X}|) + 2 \\
 &= n(H + \epsilon'),
 \end{aligned}$$

where $\epsilon' = \epsilon + \epsilon \log |\mathcal{X}| + \frac{2}{n}$.

Source coding: Zero Error, Variable length

Source coding theorem: achievability

$$R = H(X) + \epsilon'.$$

$\epsilon' = \epsilon + \epsilon \log |\mathcal{X}| + \frac{2}{n}$ can be made arbitrarily small by suitable choice of ϵ and large enough n .

We already proved earlier using Kraft inequality that $H(X)$ is the threshold rate for source coding.

Source coding: ϵ -Error, fixed length

Source coding theorem

$$\inf\{R : R \text{ is achievable}\} = H(X)$$

The code

- distinct codewords for all typical sequences: $\leq 2^{n(H+\epsilon)}$ codewords
- Another codeword for the whole non-typical set

Rate $R \leq \frac{1}{n} \log(1 + 2^{n(H+\epsilon)}) \leq H + \epsilon + \frac{1}{n}$ can be made arbitrary close to H by suitable choice of ϵ for large enough n .

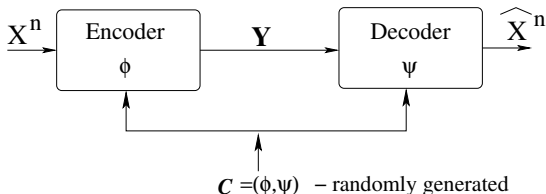
Prob. of error $\leq \Pr\{A_\epsilon^{(n)^c}\} \leq \epsilon$.

Converse to be proved later.

Source Coding: ϵ -Error, fixed length

Random coding argument

Random coding argument



Average prob. of error (avg. over C and X^n)

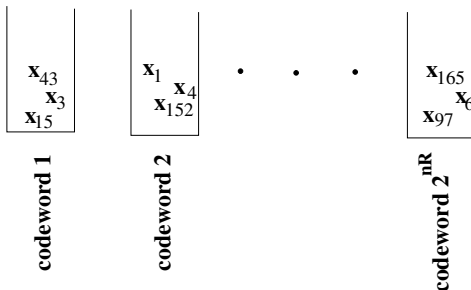
$$\begin{aligned}\bar{P}_e &= E_C [E_{X^n} P_{e,C}(X^n)] \\ &= E_{X^n} [E_C P_{e,C}(X^n)]\end{aligned}$$

For source coding, $P_{e,C}(x^n) = \mathbf{1}_{\psi(\phi(x^n)) \neq x^n}$

$$\bar{P}_e < \epsilon \Rightarrow \exists C \text{ for which } E_{X^n} P_{e,C}(X^n) < \epsilon$$

Source coding: random coding/binning

- Choose $R = H + \epsilon$, ($\epsilon > 0$)
Length of the code = $\lfloor n(H + \epsilon) \rfloor$.
- Choose any $\delta < \epsilon$.
- Consider the $2^{\lfloor n(H + \epsilon) \rfloor}$ codewords as bins.
- Put 2^n sequences in the bins randomly i.i.d. uniformly



Random coding/binning: Decoding and Prob. of error

Typicality decoding

- If there is a unique δ -typical sequence in the bin, decode that
- if there are more than one or no δ -typical sequence in the bin, declare decoding error

Error: When (1) the encoded sequence is non-typical or (2) there is another δ -typical sequence in the same bin

$$P_e^n = P_{e,1}^n + P_{e,2}^n$$

Random coding/binning: Prob. of error

$$\begin{aligned}
 P_e^n &\triangleq Pr_{\mathbf{x},c}\{\hat{\mathbf{x}} \neq \mathbf{x}\} \\
 &= \sum_{\mathcal{C}} Pr\{\mathcal{C}\} P_e(\mathcal{C}) \\
 &= \sum_{\mathbf{x}} Pr\{\mathbf{x}\} Pr_{\mathcal{C}}\{\hat{\mathbf{x}} \neq \mathbf{x}\} \\
 &\leq Pr\{A_{\delta}^{(n)c}\} \\
 &\quad + \sum_{\mathbf{x} \in A_{\delta}^{(n)}} Pr\{\mathbf{x}\} Pr_{\mathcal{C}}\{\exists \hat{\mathbf{x}} \in A_{\delta}^{(n)}, s.t., \mathbf{x} \neq \hat{\mathbf{x}}, Bin(\mathbf{x}) = Bin(\hat{\mathbf{x}})\} \\
 &= P_{e,1}^n + P_{e,2}^n
 \end{aligned} \tag{1}$$

Random coding/binning: Prob. of error

$$P_{e,1}^n$$

$$P_{e,1}^n \leq \Pr\{A_\delta^{(n)^c}\} \leq \delta.$$

This can be made arbitrarily small by a suitable choice of δ .

$$P_{e,2}^n$$

For any two δ -typical sequences, prob. that they are in the same bin $= 2^{-nR}$.

For any typical sequence, the prob. that there is another δ -typical sequence in its bin is $\leq 2^{-nR+n(H+\delta)} = 2^{-n(\epsilon-\delta)}$.

$$P_{e,2}^n \leq \Pr\{A_\delta^{(n)}\} 2^{-n(\epsilon-\delta)} \rightarrow 0$$

for $\epsilon > \delta$.

Random coding/binning: Comments

- Random binning gives universal encoding
- Typicality decoding is **not universal**
- There are universal decoding, e.g., min. entropy decoding, which also *work*
- Can prove exponential decay in the prob. of error

Source coding theorem

Converse proof

The converse: via Fano's inequality

The converse

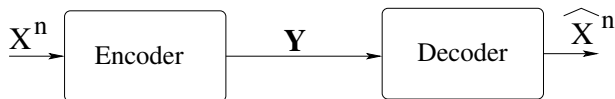
- We previously proved $H(X)$ to be the optimum average length for zero-error codes. inequality.
- We now present a converse proof for ϵ -error fixed-length codes using **Fano's inequality**.

What to prove

- If $R < H(X)$, then $P_e^{(n)} \not\rightarrow 0$ as $n \rightarrow \infty$.

The converse: via Fano's inequality

The converse



$X^n \rightarrow \mathbf{Y} \rightarrow \hat{X}^n$ is a markov chain.

If $R < H(X)$, let us assume that $R = H(X) - \epsilon$ for some $\epsilon > 0$.
Then,

$$\begin{aligned}
 H(X^n | \mathbf{Y}) &= H(X^n) - I(X^n; \mathbf{Y}) \\
 &\geq H(X^n) - H(\mathbf{Y}) \\
 &= nH(X) - nR \\
 &= n\epsilon
 \end{aligned}$$

The converse: via Fano's inequality

The converse

Now, for any sequence of codes,

$$\begin{aligned}
 P_e &\geq \frac{H(X^n|Y) - H(P_e)}{\log |\mathcal{X}|^n} \\
 &\geq \frac{n\epsilon - 1}{n \log |\mathcal{X}|} \\
 &\geq \frac{\epsilon}{\log |\mathcal{X}|} - \frac{1}{n \log |\mathcal{X}|} \\
 &\rightarrow \frac{\epsilon}{\log |\mathcal{X}|}
 \end{aligned}$$

This is a contradiction to $P_e^{(n)} \rightarrow 0$.